

Datawarehouse and Data mining

Assessment tool-3

M.oharan

CSA1610

192324115

1) Q1. Data Preprocessing in Retail Analytics

A. Handling Missing Values

- Missing: Age (C2), Monthly_Spend (C3), Purchase_Frequency (C5)
- Median Imputation: Age = 36, Spend = 41.5, Frequency = 10
- K-NN Imputation: Uses nearest similar records.

B. Outlier Detection

- Outliers: Age = 105, Monthly_Spend = 500
- Record: C4
- Treatment: Apply capping or remove the outlier.

C. Categorical Data Standardization

- Inconsistent values: M, Male, male, F, Female, FEMALE
- Standard Format:
 - Male $\rightarrow$ M, Male, male

- Female $\rightarrow$ F, Female, FEMALE

2) Q2. Covariance & Data Reduction in E-commerce Analytics

A. Covariance Analysis

- Covariance (Income & Purchase_Value): Positive
- Relationship: Higher income $\rightarrow$ Higher purchase value.
- Covariance helps identify related features.

B. Feature Selection

- Redundant Features: Income, Avg_Balance, ETI
- Selected Features: Age, Purchase_Value, Credit_Score, Orders
- Reason: Reduces redundancy and improves model efficiency.

C. Data Transformation

- Standardize all attributes except Cust_ID and Customer_Class.
- Standardization gives all features the same scale and improves model performance.

3) Q3. Data Transformation & Discretization in Education Analytics

A. Equal-Width Binning

- Study_Hours (3 bins):

2-5 6-8 9-11

- Test_Score (3 bins):

40-60 61-80 81-100

B. Equal-Frequency Binning

- Bin 1: 2, 3, 4, 5

- Bin 2: 6, 7, 8

- Bin 3: 9, 10, 11

- Test_Score:

Bin 1: 40, 50, 55, 65

Bin 2: 75, 85, 90

Bin 3: 95, 98, 100

C. Concept Hierarchy

- Study_Hours:

Low (2-4), Medium (5-7), High (8-11)

- Test_Score:

Poor (40-55), Average (65-85), Excellent

(90-100)

4)

Q4. Data Normalization

Given Data:

200, 300, 400, 600, 1000

(a) Min-Max Normalization (0-1)

Formula:

$$(x - 200) / (1000 - 200)$$

Normalized Values:

$$200 \rightarrow 0.000$$

$$300 \rightarrow 0.125$$

$$400 \rightarrow 0.250$$

$$600 \rightarrow 0.500$$

$$1000 \rightarrow 1.000$$

(b) Z-Score Normalization

$$\text{Mean} = 500$$

Standard Deviation ≈ 282.84

2-Scores:

200 $\rightarrow -1.06$

300 $\rightarrow -0.71$

400 $\rightarrow -0.35$

600 $\rightarrow 0.35$

1000 $\rightarrow 1.77$

5) Q5. Data Transformation in Online Shopping Behavior Analysis

A. Equal-Frequency Binning

- Bin 1: 15, 30, 45, 60
- Bin 2: 120, 300, 600
- Bin 3: 900, 1800, 3600
- Groups short, medium, and long visit durations.

B. 2-Score Normalization

- Mean = 747 sec
- Standard Deviation ≈ 1118 sec
- 2-score standardizes the data.
- Extreme values (1800, 3600) have high positive 2-scores.

C. Min-Max Normalization

- Formula: $(x - 15) / (3600 - 15)$
- Scales all values between 0 and 1.
- Easier to compare, but more sensitive to outliers than Z-score.